

Approximate COSMIC size to early estimate Web application development effort

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Abstract—The aim of the paper is to investigate to what extend some COSMIC-based approximate countings can be useful for project managers to early estimate effort of Web applications. We considered the number of COSMIC Functional Processes and the Average Functional Process approach proposed by the COSMIC method documentation. We carried out an empirical analysis employing data from 25 Web applications to assess whether the two approximate sizes can be exploited to get accurate effort estimations. We compared the obtained estimations with those achieved with baseline benchmarks and employing the standard COSMIC method. The analysis highlights that the considered approximate countings provide good early estimations, significantly better than those obtained with baseline benchmarks. Moreover, the first counting provides results better than the Average Functional Process approach. The use of COSMIC-based approximate countings is a suitable approach for early effort estimations.

Keywords—COSMIC, Early effort estimation, Empirical study, Linear regression

I. INTRODUCTION

Software development effort estimation concerns estimating the amount of human effort needed to realize a software system. It is crucial to have accurate effort estimations that can help project managers to plan and monitor software projects for delivering the product on time and within budget. Significant over or under-estimations can be very expensive and deleterious for a company [1]. Thus, several approaches have been proposed to carry out such an estimation. Many of them exploit an estimation model that, given data about project characteristics, are able to derive an effort estimation. One of the most important data input into these models is the application size.

Functional Size Measurement (FSM) methods have been widely applied in software engineering for estimating the size of software systems [2] [3]. In the context of Web applications, a few investigations have been performed to analyze the application of 1st generation of FSM method, i.e., Function Point Analysis (FPA) (e.g., [4] [5] [6]). Some of them have highlighted difficulties in the application of FPA suggesting the use other FSM methods for Web applications, such as COSMIC [6].

COSMIC raised the 2nd generation of FSM methods, due to several distinguishing characteristics: it was the first FSM method conform to the ISO/IEC14143/1 standard; it is based on fundamental principles of software engineering and

measurement theory; it is suitable to a wide set of different software types; and it poses the accent on the movements of data. Encouraging results in the use of COSMIC for estimating Web application development effort have been reported in [8] [9] [10] [11] [12].

In this paper we investigate to what extend some COSMIC-based approximate countings can be useful for project managers to early estimate effort of Web applications. Indeed, such approximate counting can be useful it can happen either early in the life of a project before the Functional User Requirements (FURs) have been specified in detail (“early sizing”), or when a measurement is needed but there is insufficient time or resources to apply the standard detailed method (“rapid sizing”). These motivations are not mutually exclusive and contribute to reach a trade-off between a correct measurement and time and budget available. In particular, we analyze two approximate countings. The first one is based on the use of the number of COSMIC Functional Processes while the second one is the Average Functional Process approach proposed in the COSMIC documentation [7].

We have carried out an empirical analysis employing data from 25 Web applications to assess whether the two COSMIC-based approximate sizes can be exploited to get accurate effort estimations. To this end, linear regression analysis is used to build effort estimation models based on those size measures and the accuracy of the obtained estimations is assessed by employing leave-one-out cross validation and widely used summary measures. The obtained estimations are also compared with those achieved with baseline benchmarks and the standard COSMIC method.

The rest of the paper is organized as follows. Section II reports on related work, while Section III illustrates the main aspects of COSMIC. The design of the performed empirical analysis is described in Section IV. The results of the analysis and their discussion are presented in Section V. Threats to validity are discussed in Section VI, while conclusion and future work conclude the paper.

II. RELATED WORK

Some few works have analyzed the use of COSMIC to predict the Web application development effort (e.g., [8] [9] [10] [11] [12]). To the best of our knowledge no study have been presented on the application of approximate versions

of COSMIC, i.e., Average Functional Process, Fixed Size Classification, Equal Size Bands, and Average Use Case [7]. Nevertheless, researches that can be related to our work are those proposing and/or investigating the use of approximate or simplified sizing approaches for other FSM methods.

Two approaches for simplifying the application of Function Points Analysis are the Early & Quick Function Points (EQFP) [13] and the simplified versions of NESMA [14]. The idea behind the application of EQFP is to allow a size estimation before the requirements analysis is completed. The idea of the approach is that different parts of the software can be measured at the different level of details. Thus, exploiting functional components more coarse-grained than the BFCs (i.e., EI, EO, EQ, ILF, EIF) of FPA, EQFP approach produces an approximation of the size in terms of Function Points. In particular, EQFP classifies data and processes of an application in several groups according to a hierarchy [13], where some groups contain the Base Functional Components (BFCs) of FPA and other groups are aggregation of Base Functional Processes (e.g., Macro Process, General Process) or Data Groups (e.g., Multiple Data Group).

The first approximation suggested by NESMA (named Indicative NESMA) is based on the identification of Internal Logical Files (ILF) and External Interface Files (EIF), i.e., without using the contribution of Transactions Functions (i.e., EI, EQ, and EO) of FPA, and applying predefined weights [14]. The second approximation proposed by NESMA (named Estimated NESMA) exploits all the BFCs of FPA, but the complexity of the identified components are not determined. By default, a low complexity is associated to ILF and EIF elements while a medium complexity is considered for EI, EQ, and EO.

EQFP and the two simplified NESMA approaches have been also applied to size both traditional and real-time applications for evaluating their accuracy with respect to FPA [15]. The results of the performed analysis show that EQFP produced more accurate size estimations for non real-time applications, while NESMA approaches turned out to be quite good for both applications types.

Related work can also be considered those that have analyzed the different contribution of BFCs against the overall fsu value (e.g., UFP for IFPUG [16] or CFP for COSMIC [17]) in estimating software development effort. In particular, Abran *et al.* were the first to analyze the contribution of single BFCs to the development effort of a project [18]. The results of the performed analysis reveal that correlation with development effort was different for the individual BFC types within a data set. Other two empirical studies based on ISBSG samples [19] showed that linear regression models employing two or more BFCs (e.g., EI, EO, EQ, ILF, EIF for the IFPUG method and Entry, eXit, and Read, Write for the COSMIC method) as independent variables instead of the whole single fsu value (e.g., UFP for

the IFPUG method or CFP for the COSMIC method) were characterized by more accurate estimations [20] [21]. The study presented in [22] and its replication described in [23] are further applications of the idea proposed in [20] [21], by exploiting a data set of Web applications developed by a single company operating in Italy. The results confirmed that the use of a subset of BFC types provide estimations of development effort close (i.e., no statistical significant difference) to the ones achieved the total functional size.

III. COSMIC

COSMIC is a widely adopted method for measuring software functional size and has been approved as an international standard by ISO (ISO/IEC 19761:2003) [17]. It was originally conceived for business and real-time applications, but it can be successfully applied in any other context, where software is mainly dominated by large amounts of data movements [24]. Indeed, the basic idea underlying the COSMIC approach is that, for these kinds of software, the most of development efforts are devoted to handle data movements from/to persistent storage and users. Thus, the number of these movements can provide a meaningful sight of the system size [24].

COSMIC is composed of a set of models, principles, rules, and processes applicable to the FURs of a given piece of software. The first step to obtain the COSMIC measure is to define the COSMIC Software Context Model. This model introduces the principles and concepts needed for identifying the FURs of the piece of software to be measured. Then, the COSMIC Generic Software Model is applied to the FURs to identify the components of the functionality that will be measured. In particular, the Software Context Model allows us to define why a measurement is required (Purpose) and to select the set of FURs to be included in a specific functional size measurement exercise (Scope). Furthermore, Functional User and Boundary concepts can be specified. A Functional User is a (type of) user that is a sender and/or an intended recipient of data in the FURs of a piece of software, while the Boundary is defined as a conceptual interface between the software being measured and its functional users. Then, the concepts of the Generic Software Model are applied to the FURs, to identify the set of “features” provided by the software. This model is the key of the COSMIC method, assuming that (i) each FUR can be mapped into a unique functional process, (ii) each functional process consists of sub-processes, and (iii) each sub-process may be either a data movement or a data manipulation.

A Functional Process is an elementary component of a set of FURs comprising a unique, cohesive, and independently executable set of data movements. It is triggered by a data movement from a functional user that informs the piece of software that the functional user has identified a triggering event. A Functional Sub-Process may be a data movement or a data manipulation. As an approximation for measurement

purposes, data manipulation sub-processes are not separately measured; the functionality of any data manipulation is assumed to be accounted for by the data movement with which it is associated [24]. To identify and measure data movements other three concepts have to be specified. A Triggering Event is an event (something that happens) that causes a functional user of the piece of software to initiate (trigger) one or more functional processes. A Data Group is a distinct, non empty, non ordered, and non redundant set of data attributes describing a complementary aspect of the same object of interest. A Data Attribute is the smallest piece of information, within an identified data group, carrying a meaning from the perspective of the software FURs.

Data movements can be of four types. An Entry (E) moves a data group from a functional user across the boundary into the functional process where it is required. An Exit (X) moves a data group from a functional process across the boundary to the functional user that requires it. A Read (R) moves a data group from persistent storage within each of the functional process that requires it. A Write (W) moves a data group lying inside a functional process to persistent storage.

In the measurement phase, the data movements of each functional process have to be identified and used to obtain a size measure of the software. Each data movement is counted as 1 COSMIC Function Point (CFP) that is the COSMIC measurement standard. Thus, the size of an application within a defined scope is obtained by summing the sizes of all the functional processes identified. For more details about the COSMIC method, readers are referred to the COSMIC Measurement Manual [24].

IV. EMPIRICAL ANALYSIS: DESIGN

This section presents the design of the empirical study we carried out to assess the effectiveness of two COSMIC-based approximate countings for the early estimation of Web application development effort. It is worth noting that the assessment of a size measure for effort estimation requires the use of an estimation technique to build a model relying on the selected size measure. In our empirical analysis we applied COSMIC in combination with simple linear regression (SLR), that is a model based approach, since it has been widely and successfully employed in the industrial context and in several researches to estimate development effort (see e.g., [25], [26], [27], [28]).

In our analysis we considered the number of COSMIC Functional Processes (CFunP) and the Average Functional Process (AFP) approach proposed by the COSMIC method documentation [7]. CFunP consists of determining the number of Function Processes by applying the standard COSMIC method described in Section III. AFP is the first of the four approaches suggested by the COSMIC community to obtain

early size estimations of the software¹. It consists of two steps. In the first step the functional processes of previously developed applications are identified. Then, the size for each functional process is measured in terms of COSMIC Function Points using the standard COSMIC method. In the second step, the functional processes for the new application are determined. Then, the approximate size is calculated by multiplying the number of functional processes and the average size obtained in previous step.

Thus, in the case of the first approximate counting CFunP we build a linear model based on the number of COSMIC Functional Processes, while in the case of the second approximate counting AFP we build a linear model based on the number of COSMIC Function Points.

In particular, the research questions we addressed are:

- RQ1 Can CFunP and AFP provide good early size estimations for Web effort prediction?
- RQ2 Are the early effort estimations obtained using CFunP and AFP statistically better than the estimations obtained using baseline benchmarks?
- RQ3 Are the early effort estimations obtained using CFunP and AFP comparable to the estimations obtained using the standard COSMIC method?

As done in [29] [30], we employed as baseline benchmarks the estimations obtained by employing the mean of effort (MeanEFF) and the median of effort (MedEFF) as predictor of the effort. The aim is to assess whether the estimations obtained with COSMIC Functional Processes as size measure are significant better than estimations simply based on the mean and median effort of the past projects developed by a software company. Indeed, if the estimations obtained with mean or median effort are similar to those obtained with the employed measures, then for a software company it could be more useful to simply use the mean or the median effort of the past developed applications to have an early estimation rather than dealing with more complex computations of software sizes.

A. Estimation method

We applied simple linear regression (SLR) which is a statistical technique that explores the relationship between a dependent variable and one independent variables [31], providing a prediction model described by an equation

$$y = bx + c \quad (1)$$

where y is the dependent variable, x is the independent variable, b is the coefficient that represents the amount the variable y changes when the variables x changes 1 unit, and c is the intercept. In our empirical study we exploited SLR to obtain linear regression models that use the

¹It is worth noting that the documentation the company provided us allowed to apply only this first method. We are asking further information to apply in the future the remaining three approaches

variable representing the effort as dependent and the variable denoting the number of COSMIC functional processes (or COSMIC Function Points) as independent. Once the prediction model was constructed, the effort estimation for a new Web application was obtained by sizing the application in terms of the COSMIC functional processes (or COSMIC Function Points) and using this value in the obtained model.

B. Dataset

Data for our empirical study were provided by an Italian medium-sized software company, whose core business is the development of enterprise information systems, mainly for local and central government. Among its clients, there are health organizations, research centers, industries, and other public institutions. The company is specialized in the design, development, and management of solutions for Web portals, enterprise intranet/extranet applications (such as Content Management Systems, e-commerce, work-flow managers, etc.), and Geographical Information Systems. It has about fifty employees, it is certified ISO 9001:2000, and it is also a certified partner of Microsoft, Oracle, and ESRI.

This company provided us detailed information on 25 Web applications they developed. In particular, this set includes e-government, e-banking, Web portals, and Intranet applications, developed between 2003 and 2008. All the projects have been developed by exploiting SUN J2EE or Microsoft .NET technologies. Oracle has been the most commonly adopted DBMS, but also SQL Server, Access and MySQL have been employed in some of these projects.

As for the collection of the information, the software company used timesheets to keep track of the Web application development effort. In particular, each team member annotated the information about his/her development effort every day, and weekly each project manager stored the sum of the efforts for the team. Furthermore, to collect all the significant information to calculate the values of the size measure (i.e., Functional Processes, Data movements, and so on), we prepared a questionnaire to be filled in by the project managers. All the project managers were trained on the use of the questionnaires. They took into account the COSMIC Measurement Manual (version 2.2.) [24] to apply COSMIC. One of the authors analyzed the filled templates and the analysis and design documents, in order to cross-check the provided information. The same author calculated the values of the size measure.

Table I reports on some summary statistics related to the 25 Web applications employed in our study. The variables are EFF, i.e. the actual effort, expressed in terms of person-hours, CFP, expressed in terms of number of COSMIC Function Points obtained by applying the standard COSMIC method, CFP_{afp}, expressed in terms of number of COSMIC Function Points obtained by applying the approximate counting AFP, and nCFunP, expressed in terms of number of COSMIC Functional Processes obtained by applying the

Table I
DESCRIPTIVE STATISTICS OF EFF, CFP, CFP_{afp}, AND CFUNP

Var	Obs	Min	Max	Mean	Median	Std. Dev.
EFF	25	1176	3712	2577	2686	988.14
CFP	25	163	1090	602	611	268.47
CFP _{afp}	25	183	1417	610.8	545	299.68
nCFunP	25	23	177	77	69	37.56

approximate counting CFunP. Further details on how these data have been collected are discussed in Section VI.

C. Validation method and evaluation criteria

In our analysis, we carried out a validation to verify whether or not the obtained effort predictions are useful estimations of the actual development effort. To this end, we applied a cross-validation, that is the splitting of a dataset into training and validation sets. Training sets are used to build models with SLR and validation sets are used to validate the obtained models. In particular, we exploited a leave-one-out cross validation, which means that the original dataset is divided into $n=25$ different subsets (25 is the size of the original dataset) of training and validation sets, where each validation set has one project. This validation method is widely used in empirical studies when dealing with small datasets [32].

Regarding the evaluation criteria, we decided to use several accuracy measures to assess the achieved estimations. We used some summary measures [33], namely MMRE, MdMRE, Prediction at level 25% (Pred(25)), that have been and are widely employed by researchers investigating measures and techniques for effort estimation. This can be useful to allow for the comparison with other researches that have been published in this context. Moreover, based on the observation that some criticisms have been moved to those summary measures [34], we also decided to employ Mean of Absolute Residuals (MAR) and Median of Absolute Residuals (MdAR). These measures are described in the following.

The Magnitude of Relative Error (MRE) is defined as [33]

$$MRE = \frac{|Actual - Predicted|}{Actual} \quad (2)$$

where *Predicted* and *Actual* are the predicted and the actual efforts/size, respectively. MRE has to be calculated for each observation in the test set. The MRE values are aggregated across all the observations using the mean and the median, giving rise to the Mean of MRE (MMRE), and Median of MRE (MdMRE), where the latter is less sensitive to extreme values [28]. According to [33], a good effort prediction model should have a $MMRE \leq 0.25$, to denote that the mean estimation error should be less than 25%.

The Prediction at level l is another useful indicator that measures the percentage of the estimations whose error is less than $l\%$ and l is usually set at 25%. It can be defined

as [33]

$$Pred(25) = \frac{k}{N} \quad (3)$$

N is the total number of observations and k is the number of observations whose MRE is less than or equal to 0.25. Again, according to [33], a good prediction approach should present a $Pred(25) \geq 0.75$, meaning that at least 75% of the predicted values should fall within 25% of their actual values.

MAR and MdAR are defined as the mean and the median of absolute residuals (i.e., $|Actual - Predicted|$), respectively, which are cumulative measures suggested in [34].

Moreover, we tested the statistical significance of the obtained results, by using absolute residuals, to compare prediction obtained with different approaches (e.g., to establish if COSMIC Functional Processes provided significantly better effort estimations than those achieved using MeanEFF and MedEFF) [35] [36]. In particular, we performed the T-test (and the Wilcoxon signed rank test when absolute residuals were not normally distributed) [37] to verify the following null hypothesis “the two considered population of absolute residuals have identical distributions”. This kind of test is used to verify the hypothesis that the mean of the differences in the pairs is zero.

In order to have also an indication of the practical/managerial significance of the results we analyzed the effect size [38]. Effect size is a simple way of quantifying the standardized difference between two groups. It has many advantages over using only the tests of statistical significance, since “whereas p-values reveal whether a finding is statistically significant, effect size indicates practical significance” [38]. In particular, we employed the point-biserial correlation r because it is suitable to compute the magnitude of the difference when a non parametric test (e.g., Wilcoxon) is used [39]. In the empirical software engineering field [38], the magnitude of the effect sizes measured using the point-biserial correlation is classified as follows: small (0 to 0.193), medium (0.193 to 0.456), and large (0.456 to 0.868).

V. EMPIRICAL ANALYSIS: RESULTS

We performed the linear regression analysis² to build the effort estimation models by using the dataset of 25 Web applications of Table I.

Table II shows some statistics about the linear model obtained by employing nCFunP as independent variable (i.e., the size obtained by using the approximate counting CFunP). Similarly, Table III and IV show the statistics for the models based on CFPafp (i.e., the size in terms of the number of COSMIC Function Points obtained by using the approximate

²We first verified the linear regression assumptions, i.e., the existence of a linear relationship between the independent variable and the dependent variable (linearity), the constant variance of the error terms for all the values of the independent variable (homoscedasticity), the normal distribution of the error terms (normality).

Table II
THE RESULTS OF THE LINEAR REGRESSION ANALYSIS WITH nCFunP

Variable	Value	Std.Err	t-value	p-value
Coefficient	23.66	204.83	3.69	<0.01
Intercept	755.46	2.4	9.86	<0.01

R^2	Adjusted R^2	Std. Err	F	Sign. F
0.81	0.8	441.6	97.15	<0.01

Table III
THE RESULTS OF THE LINEAR REGRESSION ANALYSIS WITH CFPafp

Variable	Value	Std. Err	t-value	p-value
Coefficient	2.96	0.31	9.69	< 0.01
Intercept	771.8	206.63	3.74	< 0.01

R^2	Adjusted R^2	Std. Err	F	Sign. F
0.803	0.79	448	94	< 0.01

counting AFP) and CFP (i.e., the size in terms of the number of COSMIC Function Points obtained by using the standard COSMIC method), respectively. A high R^2 value (and corresponding adjusted R^2 value) is an indication of the goodness of the model, since it measures the percentage of variation in the dependent variable explained by the independent variable. Other useful indicators are the F value and the corresponding p-value (denoted by Sign F), whose high and low values, respectively, denote a high degree of confidence for the prediction. Moreover, we performed a t-statistic and determined the p-value and the t-value of the coefficient and the intercept in order to evaluate its statistical significance. p-value provides the probability that a coefficient of a variable is zero, while t-value can be used to evaluate the importance of the variable for the generated model. A p-value less than 0.05 indicates that we can reject the null hypothesis and the variable is a significant predictor with a confidence of 95%. As for the t-value, a variable is significant if the corresponding t-value is greater than 1.5.

We can observe that the three models are characterized by a high R^2 value, a high F value, and a low p-value, indicating that the prediction is indeed possible with a high degree of confidence. The t-values and p-values for the corresponding coefficient and the intercept present values greater than 1.5 and less than 0.05, respectively. Moreover, the coefficient and the intercept are significant at level 0.05.

Table IV
THE RESULTS OF THE LINEAR REGRESSION ANALYSIS WITH CFP

Variable	Value	Std. Err	t-value	p-value
Coefficient	3.43	0.28	12.3	< 0.01
Intercept	512.43	183.14	2.8	0.01

R^2	Adjusted R^2	Std. Err	F	Sign. F
0.87	0.86	366.6	151.3	< 0.01

The equations of the built regression models are:

$$Effort = 23.66 * nCFunP + 755.46 \quad (4)$$

$$Effort = 2.96 * CFPafp + 771.8 \quad (5)$$

$$Effort = 3.43 * CFP + 512.43 \quad (6)$$

To evaluate the prediction accuracy of the estimation models built with the linear regression, we performed the leave-one-out cross validation, whose results are reported in Table V. We also report the results achieved by employing baseline benchmarks MeanEFF and MedEFF.

Based on the commonly accepted thresholds provided in [33], we can conclude that nCFunP and CFPafp are good indicators of Web application development effort, when used in combination with linear regression, since MMRE and MdMRE values are less than 0.25 and Pred(25) value is greater than 0.75.

Thus, we can positively answer the first research question, i.e., “*CFunP and AFP provide good early size estimations for Web effort estimation*”.

As for the comparison with the considered baseline benchmarks, we can observe that the MAR and MdAR values achieved with nCFunP and CFPafp based models are about two times less than those obtained with MeanEFF and MedEFF. Moreover, concerning the statistical significance of the results by using absolute residuals [35] [36], the Wilcoxon test revealed that the estimations obtained with nCFunP and CFPafp are significantly better than those obtained with MeanEFF and MedEFF (with p-values < 0.01). Observe that we applied the Wilcoxon test since the absolute residuals were not normally distributed as suggested by the Shapiro test (p-values < 0.01). In order to have an indication of the practical/managerial significance of this result we also analyzed the effect size and the analysis revealed a large effect size, i.e., $r=0.58$ for nCFunP vs MeanEFF and $r=0.64$ for nCFunP vs MedEFF. Similarly results were achieved for the estimations obtained in terms of CFPafp, i.e., $r=0.57$ for CFPafp vs MeanEFF and $r=0.64$ for CFPafp vs MedEFF).

Thus, we can positively answer to the second research question: i.e., *the early effort estimations obtained using CFunP and AFP are statistically better than the estimations obtained using baseline benchmarks*.

We also compared the estimations achieved with nCFunP and CFPafp. The results in terms of summary measures reported in Table V suggests that MMRE, MdMRE, and Pred(25) values are quite close. Moreover, the MAR and MdAR values obtained with nCFunP are slightly better than those achieved with CFPafp. Nevertheless, the results of the Wilcoxon test revealed that this difference is statistically significant, i.e., the absolute residuals obtained with nCFunP are significant less than those achieved with CFPafp (p-value < 0.01) with a large effect size (i.e., $r=0.52$). This is quite interesting since the CFunP is simpler than AFP since it

Table V
ACCURACY OF EFFORT ESTIMATIONS IN TERMS OF SUMMARY MEASURES

Approach	MMRE	MdMRE	Pred(25)	MAR	MdAR
nCFunP	0.17	0.15	0.8	379.79	374
CFPafp	0.19	0.17	0.8	386.6	393
CFP	0.13	0.07	0.88	275.64	198
MeanEFF	0.45	0.31	0.44	842.89	934
MedEFF	0.47	0.31	0.44	904.44	1010

does not require to determine the scaling factor used in the AFP approach to obtain the size in terms of CFP.

As for the third research question, from Table V we can observe that the MAR and MdAR values obtained with the CFP based model are less than those achieved with the nCFunP and CFPafp based models. The use of the size obtained with the standard COSMIC method also allowed us to obtain better values for the summary measures MMRE, MdMRE, and Pred(25). Furthermore, the results of the performed statistical test revealed that the difference is statistically significant (p-values = 0.02) with a medium effective size ($r=0.41$ for CFP vs nCFunP and $r=0.42$ for CFP vs CFPafp).

Thus, we cannot positively answer the third research question.

In conclusion, the analysis reported above suggests that COSMIC Functional Processes can be profitably exploited to obtain early estimations of effort for the Web applications considered in our study. In particular, both the COSMIC-based approximate countings analyzed in the paper allowed us to obtain good estimations. It is also worth to noting that the nCFunP based approach provided better estimates than the AFP based approach.

Moreover, as expected, the size obtained with the standard COSMIC method allowed us to obtain significant better effort estimations than those of the two considered approximate countings. This means that the software company involved in our study can start to employ COSMIC Functional Processes to obtain effort estimations to be employed in the management of the Web applications in the early phase of the development process. However, it has to adopt the standard COSMIC size measurement method to achieve more accurate effort estimations to be employed in the later phases of the development process.

VI. THREATS TO VALIDITY

It is widely recognized that several factors can bias the construct, internal, external, and conclusion validity of empirical studies [40] [28].

As for the construct validity, the collection of the information to determine size measures and actual effort represents a crucial aspect. Often, this is the main issue to conduct this kind of study [41]. As described in Section IV-B we have supervised the procedure employed by the involved software company to carefully collect the information we needed for

the empirical analysis. In particular, we tried to perform the data collection task in a controlled and uniform fashion. Of course we have to take into account that empirical studies do not ensure the level of confidence achieved with controlled experiments.

Some factors should be taken into account for the internal validity: subjects' authoring and designing experience, and reliability of the data and lack of standardization [28] [35] [40]. The managers involved in the study were professionals who worked in the software company. No initial selection of the subjects was carried out, so no bias has been apparently introduced. Moreover, the Web applications were developed with technologies and methods that subjects had experienced. Consequently, confounding effects from the employed methods and tools can be excluded. As for the reliability of the data and lack of standardization, the questionnaires used were the same for all the Web applications and the project managers were instructed on how to use the questionnaires, to correctly provide the required information. Instrumentation effects in general did not occur in this kind of studies. As for the conclusion validity we carefully applied the estimation methods and the statistical tests, verifying all the required assumptions.

With regard to the external validity, the applications involved in our empirical analysis are representative samples of modern Web applications, taking into account their type, functionality, target platforms, and complexity. Thus, we are confident that the type of analyzed Web applications did not bias the validity of the achieved results. On the other hand, it is recognized that the results obtained in an industrial context might not hold in other contexts. Indeed, each context might be characterized by some specific project and human factors, such as development process, developer experience, application domain, tools, technologies used, time, and budget constraints [42].

VII. CONCLUSION

The empirical analysis have highlighted that the considered COSMIC-based approximate countings provide good early estimations, significantly better than those obtained with baseline benchmarks. Moreover, the first counting provides results better than the Average Functional Process approach. Thus, we can conclude that the use of COSMIC-based approximate countings resulted to be a suitable approach for early effort estimations. As expected the estimations obtained with the approximate sizes are worse than those achieved employing the size resulted from the application of the standard COSMIC method.

Since replications of a study, in different setting and with larger datasets, are always required in this field, as future work, we plan to collect and analyze data from other companies. Furthermore, we intend to investigate the other approximate sizing approaches proposed in the COSMIC documentation [7]. In particular, we would like to analyze

pros and cons of the different approaches not only with respect to the accuracy of the estimation but also with respect to the information required to apply the methods and then the time needed to carry out the measurement.

We are also planning to investigate whether the kind of applications considered can influence the obtained estimation models.

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